

Types of Models

DAEN 459 Fall 2025

1. Process Models

- **Flowcharts:** Visual representations of workflows, showing the sequence of steps and decision points.
- **Business Process Model and Notation (BPMN):** A more formal representation for business processes, useful for complex or standardized operations.
- **Value Stream Maps:** Diagrams that capture the flow of materials and information in a process, highlighting areas of value and waste.

2. System Architecture Models

- **Block Diagrams:** High-level representations of system components and their interactions.
- **Functional Flow Block Diagrams (FFBD):** Displays functions of a system in sequence, helpful in defining operations and dependencies.
- **System Context Diagrams:** Shows the interactions between a system and its external environment, including users and other systems.

3. Data Flow Models

- **Data Flow Diagrams (DFD):** Focuses on the flow of data between processes, data stores, and external entities.
- **Entity-Relationship Diagrams (ERD):** Visualizes relationships between data entities, often used for database design.
- **Unified Modeling Language (UML) Diagrams:** Diagrams like sequence or use case diagrams can model data flow or interactions between components.

4. Simulation Models

- **Discrete Event Simulation (DES):** Models systems as a sequence of events over time, useful for analyzing operational processes.
- **Agent-Based Models (ABM):** Models the behavior of individual agents and their interactions within a system.
- **Monte Carlo Simulations:** Uses random sampling and statistical modeling to predict the impact of uncertainty in systems.

5. Physical Models

- **CAD Models (Computer-Aided Design):** Useful for representing physical components and assemblies.
- **Prototype Models:** Early versions of a physical or software system to visualize and test system elements.

6. Network Models

- **PERT/CPM Diagrams (Program Evaluation Review Technique / Critical Path Method):** Useful in project management for scheduling and identifying the longest sequence of tasks.
- **Network Topology Diagrams:** Visualizes the structure of a communication network or a supply chain.

7. Performance Models

- **Queueing Models:** Used to represent systems where resources are shared (e.g., production lines, customer service), helpful in analyzing bottlenecks.
- **Markov Chains:** Models the probabilities of moving between different states in a system over time, useful for reliability analysis.

8. Economic and Financial Models

- **Cost-Benefit Analysis:** Simple models comparing the costs and benefits of a system or process.
- **Break-Even Analysis:** Helps determine the point at which total costs and total revenues are equal.

9. Control Models

- **State Machine Diagrams:** Models the states of a system and transitions between them, useful in control systems or software design.
- **Feedback Loop Models:** Represent how outputs are fed back into a system as inputs, commonly used in systems engineering and control theory.

10. Operational Models

- **Gantt Charts:** For scheduling and tracking tasks over time in project management.
- **Workforce Models:** Represents resource allocation, including human capital, to understand workflow efficiency.

11. Stakeholder Interaction Models

- **Use Case Diagrams:** Represent how different actors (users or systems) interact with the system.
- **Stakeholder Maps:** Visualizes relationships and influences of different stakeholders in a system or process.